

CS-007 Wet Layup & Vacuum Bagging

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Title	Wet Layup & Vacuum Bagging
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Supersedes	Verbal wet-layup training

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted w/ Claude — SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial outline

1. Purpose

Wet layup is our second core process (seat, flat panels, foam-wrapped wing elements). In SN5 the knowledge was verbal plus one Slack SOP; this doc writes it down with the numbers attached.

2. Scope

Hand lamination with West System 105, vacuum-bagged, on molds (CS-004-prepped), flat plates, and foam cores (foam-wrapped wing elements). Infusion: CS-006.

3. References

Ref	Document	Where
R1	West 105/206 TDS (5:1 by volume; pot life 20–25 min @72 °F; thin-film cure 10–15 h)	04 Datasheets/WEST_SYSTEM_105_Epoxy_Resin_wit
R2	West 105/205 TDS (fast: pot life 9–12 min)	04 Datasheets/WestSystems-105-205-Epoxy-Resin
R3	VB160 film, PP180 peel ply, breather	04 Datasheets/
R4	CS-002 (stack), CS-004 (mold prep), CS-008 (mixing/exotherm)	this series

Ref	Document	Where
R5	Simon’s layup SOP (Slack, 2026-05-30): 105 “big tank, blue label” + 206 “large metal can, red label”, 5:1	Slack record

4. Definitions

Breather (fluffy bleed cloth — never touching part edges it could bond to), bridging, wet-out, 60:40 fiber:resin target **by weight**.

5. Equipment & Materials

West 105 + 206 (default) or 205 (small parts), calibrated pumps (5:1 by volume, R1) or scale (weight ratios in R1: 5.36:1 for 206), brushes/rollers/squeegees, peel ply, breather, VB160, tacky tape, vacuum pump + trap + gauge.

6. Safety

Per West SDS + team rules: nitrile gloves (sensitizer — same warning as CS-004 §6), ventilation, no solvent on skin; exotherm rules per CS-008 (105/205 in a pot gets hot fast). Respirator with A-filter for large overhead/enclosed laminating.

7. Procedure (outline — expand with photos on first parts)

1. Prerequisites: stack frozen (CS-002 §7.2), mold released (CS-004), cloth cut + weighed.
2. Resin math: mix fiber mass $\times 0.9$ for hand layup. (Yes, that is more than the 60:40 target implies — the squeegee and the breather take the excess out; the *cured part* lands near 60:40. Record mixed vs final masses on the WO to tune this factor.)
3. Mix per CS-008: 5:1 by volume with the calibrated pumps (R1) — one FULL pump stroke each, never partial strokes; pot life 206 @72 °F: 20–25 min per 100 g pot.
4. Wet out ply-by-ply per the WO ply table; squeegee each ply — shiny-wet, not swimming.

[PHOTO: correctly wetted ply — uniform sheen, weave visible, no pooling]

5. Lay peel ply over the full laminate.
6. Lay breather over the peel ply, kept clear of part edges it could bond to.
7. Bag with VB160 + tacky tape, adding pleats at corners (CS-006 §7.3 rules apply).
8. Pull vacuum to **25 inHg**; drop test: isolate pump, **2 inHg over 10 minutes** before walking away. Record readings on the WO.

[PHOTO: bagged wet layup under vacuum, pleats visible, gauge in frame]

9. Cure per R1 (206: thin-film solid 10–15 h @72 °F; leave bagged overnight minimum). Record times/masses on the WO.
10. Foam-wrapped elements (RW E2–E5 class): same flow over a shaped foam core; watch vacuum crush on light foams — reduce vacuum if the core dents (record the level that worked per part on the WO).

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Vacuum	25 inHg; drop 2 inHg/10 min	Gauge	WO
Wet-out	No white/dry weave visible	Visual per ply	WO
Mass	WO target $\pm 10\%$	Scale	WO qualityChecks

9. Records

Same as CS-006 §9: times, masses, gauge readings, photos into the WO.